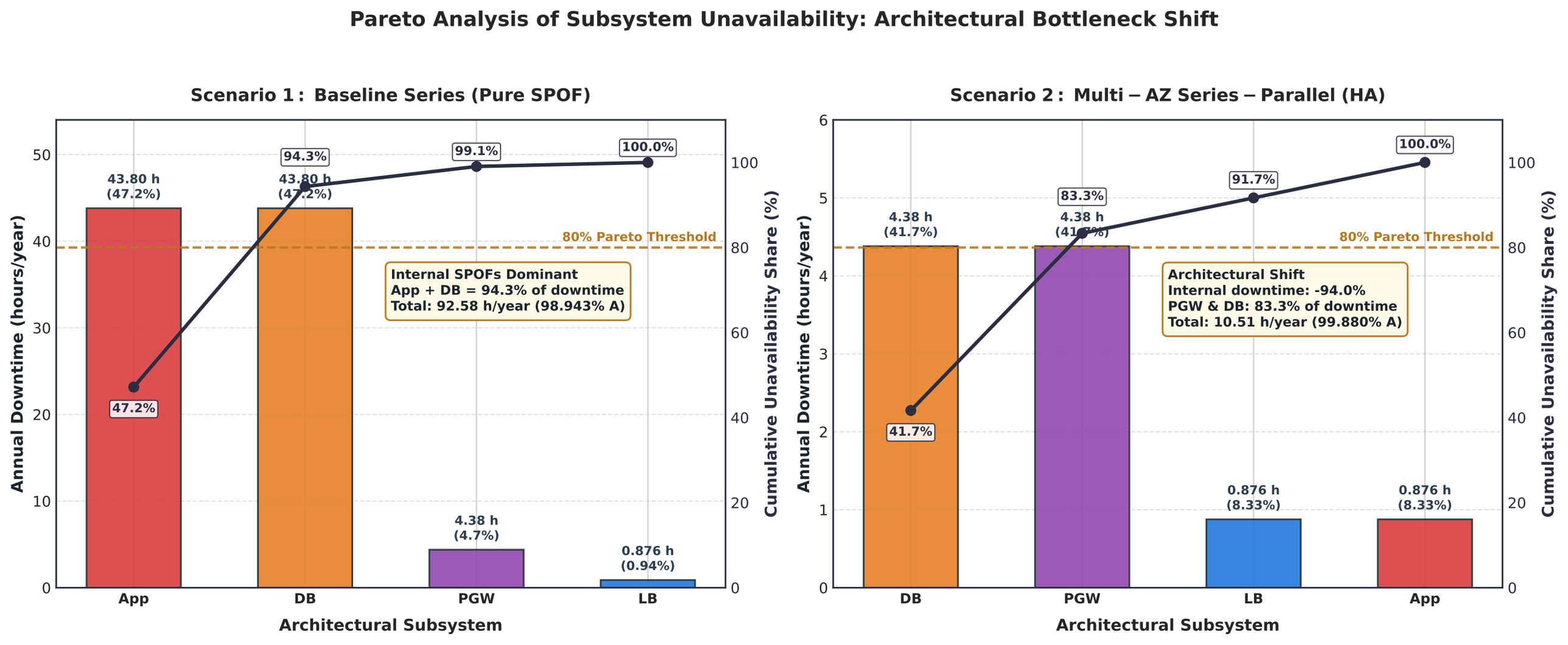
Pareto Analysis of Subsystem Unavailability: Architectural Bottleneck Shift
Scenario 1 : Baseline Series (Pure SPOF)
Annual Downtime (hours/year)
Cumulative Unavailability Share (%)
Architectural Subsystem
App
DB
PGW
LB
43.80 h
(47.2%)
43.80 h
(47.2%)
4.38 h
(4.7%)
0.876 h
(0.94%)
47.2%
94.3%
99.1%
100.0%
80% Pareto Threshold
Internal SPOFs Dominant
App + DB = 94.3% of downtime
Total: 92.58 h/year (98.943% A)
Scenario 2 : Multi – AZ Series – Parallel (HA)
Annual Downtime (hours/year)
Cumulative Unavailability Share (%)
Architectural Subsystem
DB
PGW
LB
App
4.38 h
(41.7%)
4.38 h
(41.7%)
0.876 h
(8.33%)
0.876 h
(8.33%)
41.7%
83.3%
91.7%
100.0%
80% Pareto Threshold
Architectural Shift
Internal downtime: -94.0%
PGW & DB: 83.3% of downtime
Total: 10.51 h/year (99.880% A)